
AP Physics 1: Algebra-Based Practice Exam

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AP[®] PHYSICS 1 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N·m ² /C ²
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Universal gravitational constant, $G = 6.67 \times 10^{-11}$ m ³ /kg·s ²
Speed of light, $c = 3.00 \times 10^8$ m/s	Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²

UNIT SYMBOLS	meter, m	kelvin, K	watt, W	degree Celsius, °C
	kilogram, kg	hertz, Hz	coulomb, C	
	second, s	newton, N	volt, V	
	ampere, A	joule, J	ohm, Ω	

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam.

- I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- II. Assume air resistance is negligible unless otherwise stated.
- III. In all situations, positive work is defined as work done on a system.
- IV. The direction of current is conventional current: the direction in which positive charge would drift.
- V. Assume all batteries and meters are ideal unless otherwise stated.

AP[®] PHYSICS 1 EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_n|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$K = \frac{1}{2} m v^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$x = A \cos(2\pi f t)$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$L = I \omega$$

$$\Delta L = \tau \Delta t$$

$$K = \frac{1}{2} I \omega^2$$

$$|\vec{F}_s| = k |\vec{x}|$$

$$U_s = \frac{1}{2} k x^2$$

$$\rho = \frac{m}{V}$$

a = acceleration
 A = amplitude
 d = distance
 E = energy
 f = frequency
 F = force
 I = rotational inertia
 K = kinetic energy
 k = spring constant
 L = angular momentum
 ℓ = length
 m = mass
 P = power
 p = momentum
 r = radius or separation
 T = period
 t = time
 U = potential energy
 V = volume
 v = speed
 W = work done on a system
 x = position
 y = height
 α = angular acceleration
 μ = coefficient of friction
 θ = angle
 ρ = density
 τ = torque
 ω = angular speed

$$\Delta U_g = m g \Delta y$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$\vec{g} = \frac{\vec{F}_g}{m}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I \Delta V$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

A = area
 F = force
 I = current
 ℓ = length
 P = power
 q = charge
 R = resistance
 r = separation
 t = time
 V = electric potential
 ρ = resistivity

WAVES

$$\lambda = \frac{v}{f}$$

f = frequency
 v = speed
 λ = wavelength

GEOMETRY AND TRIGONOMETRY

Rectangle
 $A = bh$
 A = area
 C = circumference
 V = volume

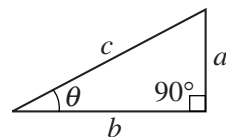
Triangle
 $A = \frac{1}{2} bh$
 S = surface area
 b = base
 h = height
 ℓ = length

Circle
 $A = \pi r^2$
 $C = 2\pi r$
 w = width
 r = radius

Rectangular solid
 $V = \ell wh$
Right triangle
 $c^2 = a^2 + b^2$

Cylinder
 $V = \pi r^2 \ell$
 $S = 2\pi r \ell + 2\pi r^2$
 $\sin \theta = \frac{a}{c}$
 $\cos \theta = \frac{b}{c}$
 $\tan \theta = \frac{a}{b}$

Sphere
 $V = \frac{4}{3} \pi r^3$
 $S = 4\pi r^2$



PHYSICS 1

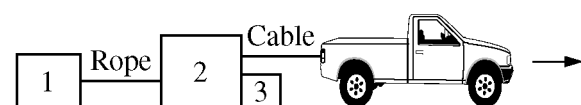
Section I

Time—1 hour and 30 minutes

40 Questions

Note: To simplify calculations, you may use $g = 10 \text{ m/s}^2$ in all problems.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



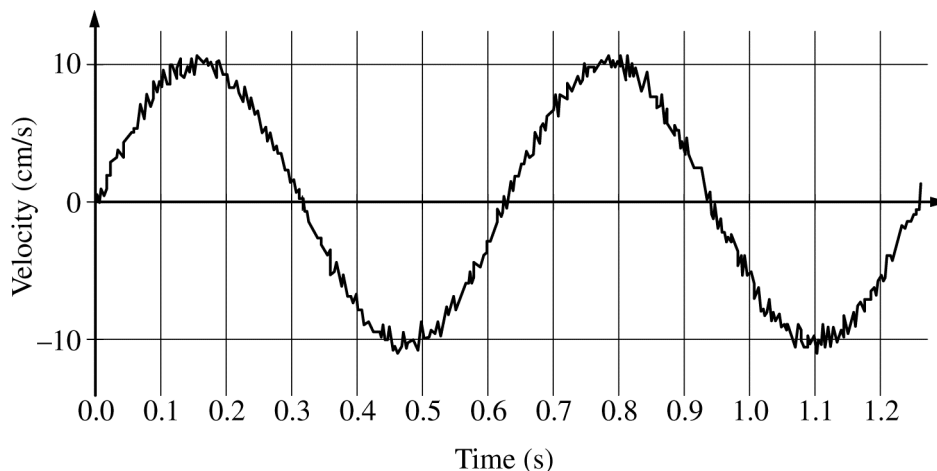
1. The figure above shows a truck pulling three crates across a rough road. Which of the following shows the directions of all the horizontal forces acting on crate 2?

- (A)
- (B)
- (C)
- (D)

2. Two blocks are on a horizontal, frictionless surface. Block A is moving with an initial velocity of v_0 toward block B, which is stationary, as shown above. The two blocks collide, stick together, and move off with a velocity of $v_0/3$. Which block, if either, has the greater mass?

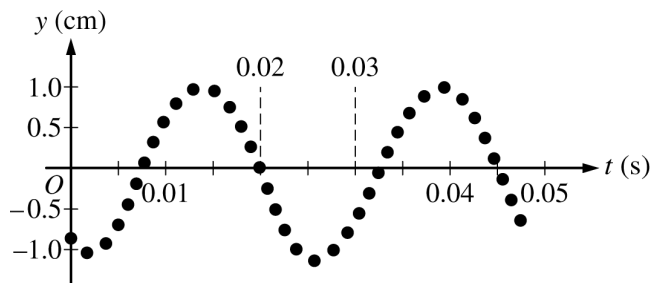
- (A) Block A
 (B) Block B
 (C) Neither; their masses are the same.
 (D) The answer cannot be determined without knowing the mass of one of the blocks.

Questions 3-5 refer to the following material.



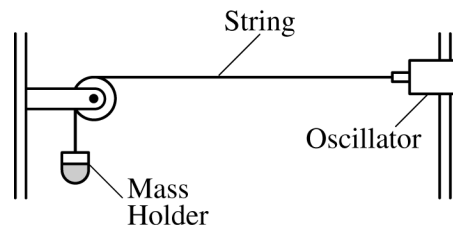
A student sets an object attached to a spring into oscillatory motion and uses a motion detector to record the velocity of the object as a function of time. A portion of the recorded data is shown in the figure above.

3. The total change in the object's speed between 1.0 s and 1.1 s is most nearly
- (A) zero
 - (B) 5 cm/s
 - (C) 10 cm/s
 - (D) 15 cm/s
4. The acceleration of the object at time $t = 0.7$ s is most nearly equal to which of the following?
- (A) The value of the graph where it crosses the 0.7 s grid line
 - (B) The slope of the line connecting the origin and the point where the graph crosses the 0.7 s grid line
 - (C) The area under the curve between where the graph crosses the time axis near 0.63 s and time 0.7 s
 - (D) The slope of the tangent to a best-fit sinusoidal curve at 0.7 s
5. The frequency of oscillation is most nearly
- (A) 0.63 Hz
 - (B) 0.80 Hz
 - (C) 1.25 Hz
 - (D) 1.60 Hz

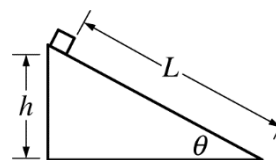


6. One end of a string is fixed in place and the other end is vibrated, allowing a wave to propagate along the string. The vertical position y of a single point on the string is measured at fixed intervals of time t as the wave passes. The graph above shows y as a function of t for the point. A separate measurement indicates that the wave speed on the string is 12 m/s . The wavelength of the wave is most nearly

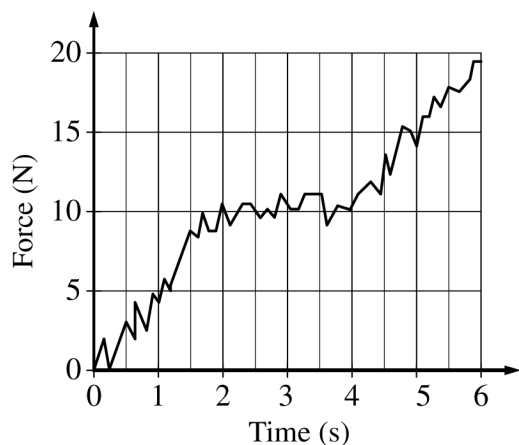
(A) 0.0021 m
 (B) 0.15 m
 (C) 0.30 m
 (D) 480 m



7. Using the setup above, a student tests how the tension in a string affects the frequency of the fundamental standing wave (with one antinode). In each trial, with a known mass in the holder, the student adjusts the oscillator frequency to produce the fundamental standing wave. The student notices that as the mass in the holder increases, the string stretches and the total string length, from the oscillator to the mass holder, increases. Which of the following procedures could help the student decide if the string's total length has a separate effect from the tension on the fundamental frequency?
- (A) Repeating the experiment with a string that does not noticeably stretch when the tension is increased
 (B) Repeating the experiment but producing standing waves with two antinodes
 (C) Repeating the experiment using a different distance between the oscillator and pulley
 (D) Repeating the experiment using lighter blocks

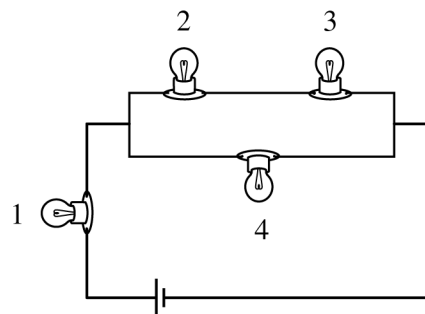


8. A box of mass m is initially at rest at the top of a ramp that is at an angle θ with the horizontal. The block is at a height h and length L from the bottom of the ramp. The block is released and slides down the ramp. The coefficient of kinetic friction between the block and the ramp is μ . What is the kinetic energy of the box at the bottom of the ramp?
- (A) mgh
 (B) $\mu mgL \cos \theta$
 (C) $mgh - \mu mgL \cos \theta$
 (D) $mgL - \mu mgh \cos \theta$



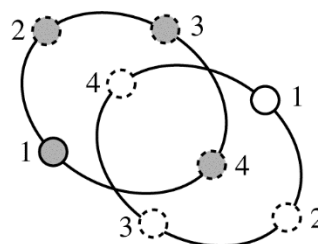
9. During an experiment a student records the net horizontal force exerted on an object moving in a straight line along a horizontal frictionless track. The graph above shows the force as a function of time. Of the following, which is the best approximation of the magnitude of the change in momentum of the object between 0 s and 4 s?

(A) 20 kg•m/s
 (B) 30 kg•m/s
 (C) 40 kg•m/s
 (D) The magnitude of the change in momentum cannot be determined without knowing the mass of the object.



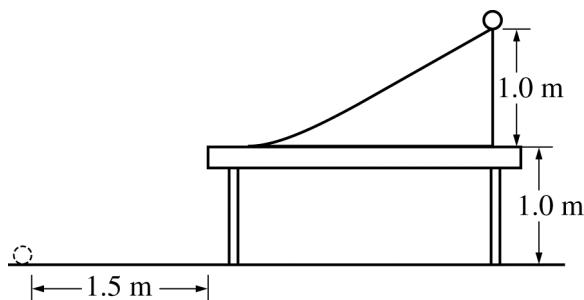
10. The figure above shows four identical lightbulbs connected to a battery. Which of the following correctly ranks the power P dissipated by the lightbulbs?

(A) $P_1 > (P_2 = P_3 = P_4)$
 (B) $(P_2 = P_3 = P_4) > P_1$
 (C) $P_1 > P_4 > (P_2 = P_3)$
 (D) $P_1 = P_2 = P_3 = P_4$



11. The figure above shows the paths of two stars of equal mass as they orbit their common center of mass. The positions of the stars at four different times are labeled in the figure. At which of the positions do the stars have their greatest speed?
- (A) Position 2 only
 (B) Position 4 only
 (C) Positions 1 and 3 only
 (D) All of the positions, since the stars move with constant speed

Questions 12-14 refer to the following material.



A cylinder at rest is released from the top of a ramp, as shown above. The ramp is 1.0 m high, and the cylinder rolls down the ramp without slipping. At the bottom of the ramp, the cylinder makes a smooth transition to a small section of a horizontal table and then travels over the edge at a height of 1.0 m above the floor, eventually landing on the floor at a horizontal distance of 1.5 m from the table.

12. As the cylinder rolls down the ramp, how do the potential energy of the cylinder-Earth system and the kinetic energy of the cylinder change, if at all?

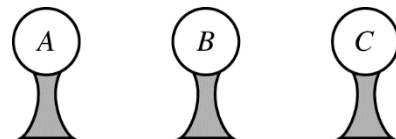
Potential Energy of Cylinder-Earth System	Kinetic Energy of Cylinder
(A) Stays the same	Increases
(B) Stays the same	Decreases
(C) Decreases	Increases
(D) Decreases	Decreases

13. After the cylinder leaves the table, but before it lands, how do the rotational kinetic energy and translational kinetic energy of the cylinder change, if at all?

Rotational Kinetic Energy	Translational Kinetic Energy
(A) Increases	Increases
(B) Increases	Stays the same
(C) Stays the same	Increases
(D) Stays the same	Stays the same

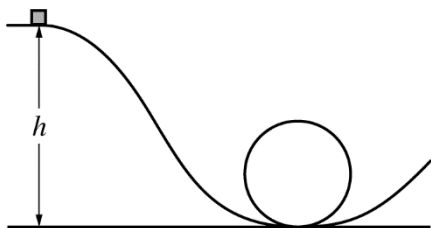
14. A sphere with the same mass and radius as the original cylinder, but a smaller rotational inertia, is released from rest from the top of the ramp. K_s and K_c are the sphere's and the cylinder's total kinetic energy at the bottom of the ramp, respectively. How do K_s and K_c compare, and why?

- (A) $K_s < K_c$, because the sphere will gain less rotational kinetic energy.
 (B) $K_s < K_c$, because the sphere has a greater acceleration and therefore has less time to gain kinetic energy.
 (C) $K_s = K_c$, because both objects accelerate at the same rate.
 (D) $K_s = K_c$, because the gravitational force does equal work on each object as it rolls down the ramp.



15. Three identical metal spheres on insulating stands are initially isolated from each other, as shown above. The spheres A, B, and C carry electric charges $+12Q$, $-12Q$, and $+12Q$, respectively. Whenever two of the spheres come into contact, each sphere ends up with the same net charge as the other sphere. Spheres A and B are brought into contact with each other and separated. Spheres A and C are then brought into contact with each other and separated. Which of the following correctly shows the final charge on each sphere?

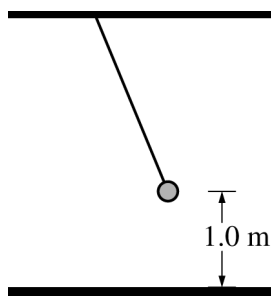
	<u>A</u>	<u>B</u>	<u>C</u>
(A)	0	0	$+12Q$
(B)	$+4Q$	$+4Q$	$+4Q$
(C)	$+6Q$	0	$+6Q$
(D)	$+12Q$	$+12Q$	$-12Q$



Note: Figure not drawn to scale.

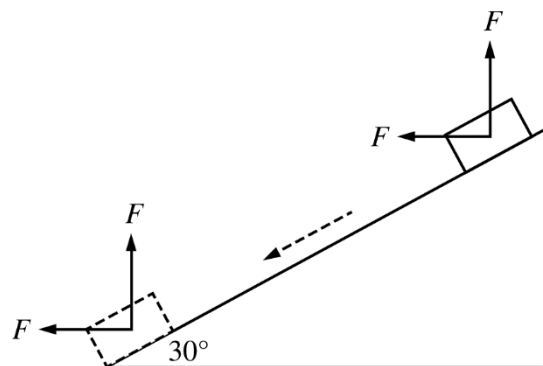
16. A small block slides without friction along a track toward a circular loop. The block has more than enough speed to remain firmly in contact with the track as it goes around the loop. The magnitude of the block's acceleration at the top of the loop is

(A) zero
 (B) greater than zero but less than g
 (C) equal to g
 (D) greater than g



17. A 0.5 kg pendulum bob is raised to 1.0 m above the floor, as shown in the figure above. The bob is then released from rest. When the bob is 0.8 m above the floor, its speed is most nearly

(A) 5 m/s
 (B) 4 m/s
 (C) 2 m/s
 (D) 1 m/s



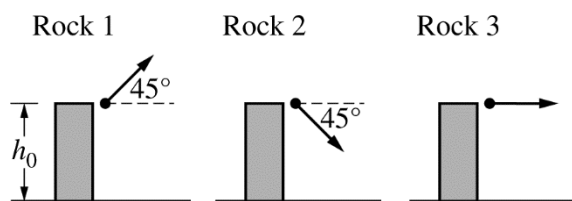
18. A block is released from rest and slides down a frictionless ramp inclined at 30° from the horizontal. When the block reaches the bottom, the block-Earth system has mechanical energy E_0 . The experiment is repeated, but now horizontal and vertical forces of magnitude F are exerted on the block while it slides, as shown above. When the block reaches the bottom, the mechanical energy of the block-Earth system

(A) is greater than E_0
 (B) is equal to E_0
 (C) is less than E_0
 (D) cannot be determined without knowing F

19. An apple is released from rest 500 m above the ground. Due to the combined forces of air resistance and gravity, it has a speed of 40 m/s when it reaches the ground. What percentage of the initial mechanical energy of the apple-Earth system was dissipated due to air resistance? Take the potential energy of the apple-Earth system to be zero when the apple reaches the ground.

(A) 16%
 (B) 40%
 (C) 60%
 (D) 84%

Questions 20-21 refer to the following material.



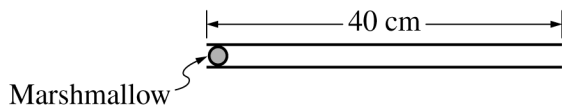
Three identical rocks are launched with identical speeds from the top of a platform of height h_0 . The rocks are launched in the directions indicated above.

20. Which of the following correctly relates the magnitude v_y of the vertical component of the velocity of each rock immediately before it hits the ground?

- (A) $(v_{y1} = v_{y2}) > v_{y3}$
- (B) $v_{y1} > v_{y3} > v_{y2}$
- (C) $v_{y2} > v_{y3} > v_{y1}$
- (D) $v_{y1} = v_{y2} = v_{y3}$

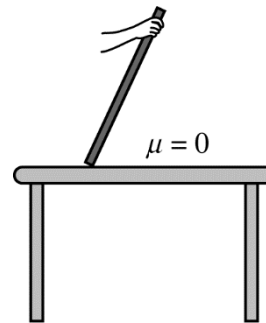
21. Rock 1, of mass m , reaches a maximum height $h_{\max}$ after being launched. During the time between the instant rock 1 is launched from height h_0 and the instant it returns to height h_0 , the work done on the rock by the gravitational force is

- (A) 0
- (B) $-mgh_0$
- (C) $-mg(h_{\max} - h_0)$
- (D) $2mg(h_{\max} - h_0)$

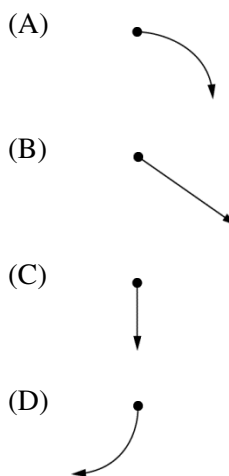


22. A 2.5 g marshmallow is placed in one end of a 40 cm pipe, as shown in the figure above. A person blows into the left end of the pipe to eject the marshmallow from the right end. The average net force exerted on the marshmallow while it is in the pipe is 0.7 N. The speed of the marshmallow as it leaves the pipe is most nearly

(A) 4.7 m/s
 (B) 11 m/s
 (C) 15 m/s
 (D) 280 m/s



23. A meterstick is held as shown above and then released from rest. The tabletop has negligible friction. Which figure below best indicates the path of the center of mass of the meterstick as it falls?



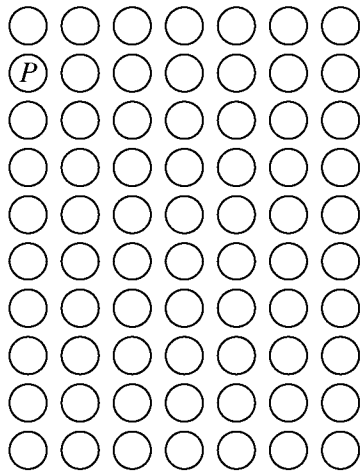


Figure 1

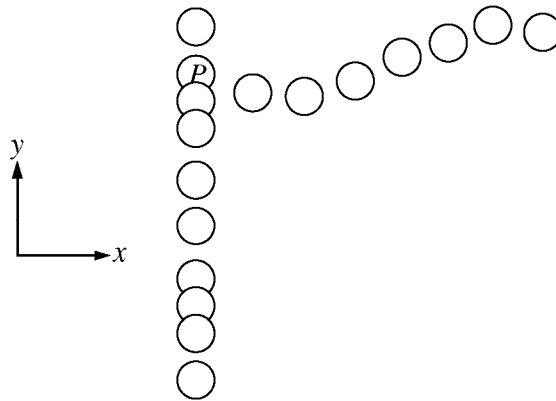
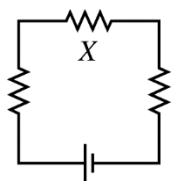


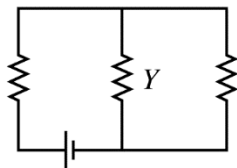
Figure 2

24. Figure 1 above represents atoms in a solid material. A particle hits atom P and the atom vibrates, disturbing nearby atoms. Figure 2 represents just the vertical and horizontal rows containing atom P and shows the propagation of the disturbance along them. Which of the following statements correctly indicates the direction of atom P 's vibration and the type of wave along each row?

<u>P's Vibration</u>	<u>Wave Along Vertical Row</u>	<u>Wave Along Horizontal Row</u>
(A) Along y axis	Longitudinal	Transverse
(B) Along y axis	Transverse	Longitudinal
(C) Along x axis	Longitudinal	Transverse
(D) Along x axis	Transverse	Longitudinal

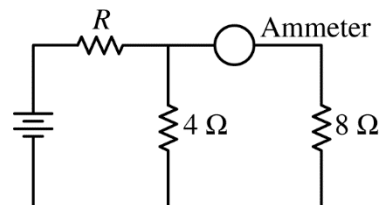


Circuit 1

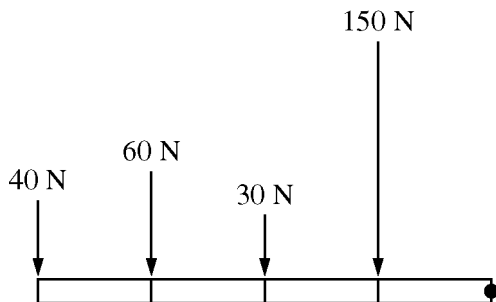


Circuit 2

25. The two circuits shown above contain identical resistors and batteries. Which of the following best explains why resistors X and Y dissipate the same power?
- (A) Resistors X and Y have the same potential difference across them because each is the second resistor in its circuit.
 - (B) Both circuits have identical batteries and contain the same number of identical resistors.
 - (C) Although each circuit has a different total current, the current through an individual resistor depends only on its resistance.
 - (D) Compared to circuit 1, the total current through circuit 2 is twice as much, but only half that current flows through resistor Y .

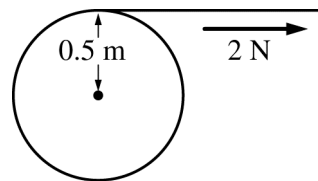


26. In the circuit above, the current through the ammeter is 2 A. What is the current through the resistor of unknown resistance R ?
- (A) 6 A
 - (B) 4 A
 - (C) 3 A
 - (D) The current cannot be determined without knowing the value of R .



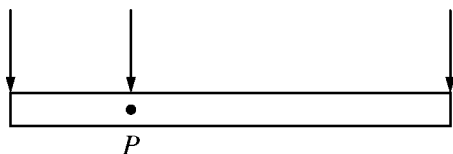
27. The figure above represents a stick of uniform density that is attached to a pivot at the right end and has equally spaced marks along its length. Any one of the four forces shown can be exerted on the stick as indicated. Which force will create the largest rate of change in the stick's angular momentum?

(A) The 30 N force
 (B) The 40 N force
 (C) The 60 N force
 (D) The 150 N force



28. A disk with radius of 0.5 m is free to rotate around its center without friction. A string wrapped around the disk is pulled, as shown above, exerting a 2 N force tangent to the edge of the disk for 1 s. If the disk starts from rest, what is its angular speed after 1 s?

(A) 0 rad/s
 (B) 1 rad/s
 (C) 4 rad/s
 (D) It cannot be determined without knowing the rotational inertia of the disk.



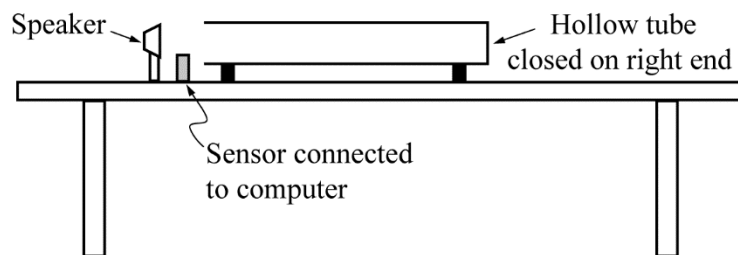
Top View

29. The figure above shows a rod that is fixed to a horizontal surface at pivot P . The rod is initially rotating without friction in the counterclockwise direction. At time t , three forces of equal magnitude are applied to the rod as shown. Which of the following is true about the angular speed and direction of rotation of the rod immediately after time t ?

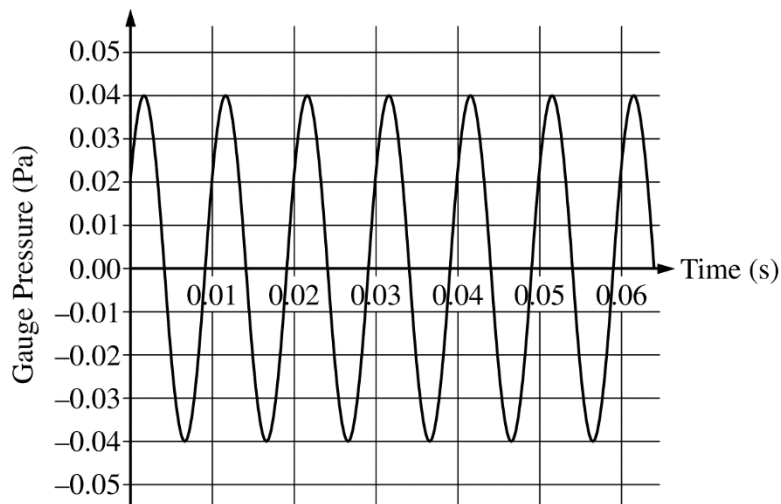
	<u>Angular Speed</u>	<u>Direction of Rotation</u>
(A)	Decreasing	Counterclockwise
(B)	Decreasing	Clockwise
(C)	Increasing	Counterclockwise
(D)	Increasing	Clockwise

30. Two systems are in oscillation: a simple pendulum swinging back and forth through a very small angle and a block oscillating on a spring. The block-spring system takes twice as much time as the pendulum to complete one oscillation. Which of the following changes could make the two systems oscillate with the same period?
- (A) Increasing the mass of the pendulum bob
 (B) Increasing the angle through which the pendulum swings by a small amount
 (C) Decreasing the mass of both the block and the pendulum bob
 (D) Shortening the pendulum

Questions 31-32 refer to the following material.



Students use a speaker connected to a sine wave generator to project sound waves into the open left end of a tube that is closed on the right end. The speaker is very close to the tube. A sensor is used to measure the pressure differences created by the sound and the results are displayed on a computer. The entire apparatus is in a sealed chamber that can be filled with different gases.

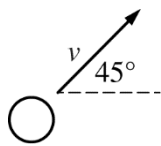


31. The students vary the frequency of the sound until the maximum loudness from the tube is achieved. The graph above represents the computer display of gauge pressure (the amount by which the pressure is above or below atmospheric pressure) as a function of time. What are the amplitude and frequency of the wave?

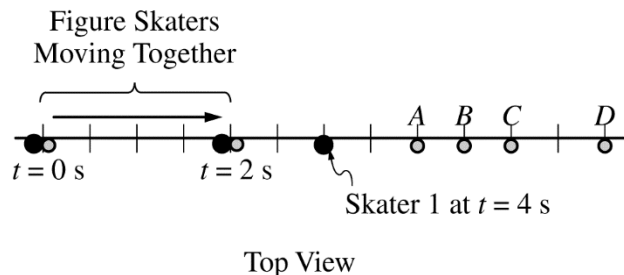
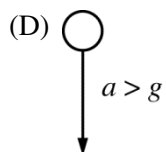
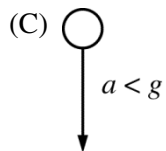
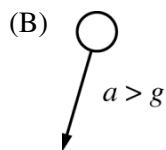
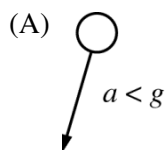
	<u>Amplitude</u>	<u>Frequency</u>
--	------------------	------------------

- | | | |
|-----|---------|---------|
| (A) | 0.04 Pa | 0.01 Hz |
| (B) | 0.04 Pa | 100 Hz |
| (C) | 0.08 Pa | 0.01 Hz |
| (D) | 0.08 Pa | 100 Hz |
32. The chamber is now filled with a gas that has an unknown speed of sound. The students find that the lowest frequency at which they can hear a maximum in the sound level is 200 Hz. If the students slowly increase the frequency, what is the next frequency at which the sensor would measure a maximum in the sound level?

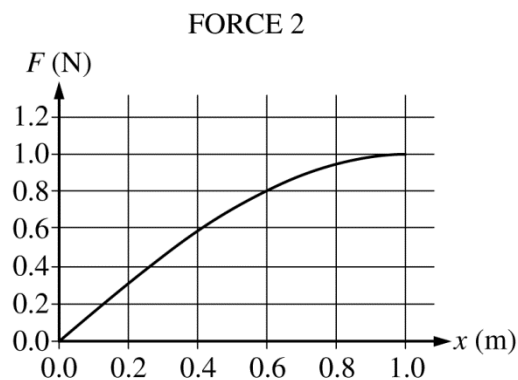
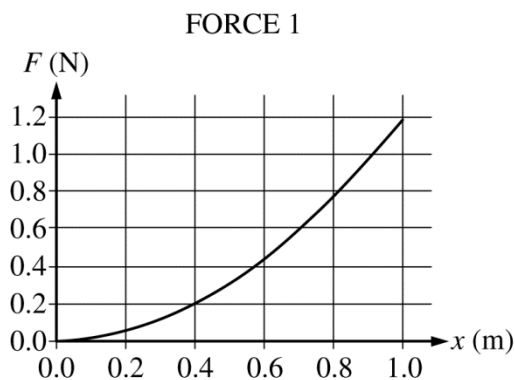
- (A) 300 Hz
 (B) 400 Hz
 (C) 600 Hz
 (D) 800 Hz



33. A hollow plastic ball is projected into the air. There is significant air resistance opposing the ball's motion, so the magnitude of the ball's acceleration is not equal to g . At time t , the ball is moving up and to the right at an angle of 45° to the horizontal, as shown above. Which of the following best shows the magnitude a and the direction of the ball's acceleration at time t ?



34. At time $t = 0$ two figure skaters are moving together over ice with negligible friction, as shown above. Skater 1, represented by the large black dot, is twice as massive as skater 2, represented by the gray dot. At $t = 2$ s the skaters push off of one another. The location of skater 1 is shown at $t = 4$ s. At $t = 4$ s, skater 2 is located at which of the labeled points?
- (A) Point A
(B) Point B
(C) Point C
(D) Point D
35. A charged sphere located at the origin of the xy -plane exerts a force of magnitude F on a charged particle located at $(+d, 0)$. The sphere will exert a force $F/2$ on the particle if the particle is moved to which of the following locations?
- (A) $(+2d, 0)$
(B) $(+d, +d)$
(C) $(+d/2, 0)$
(D) $(0, -d)$

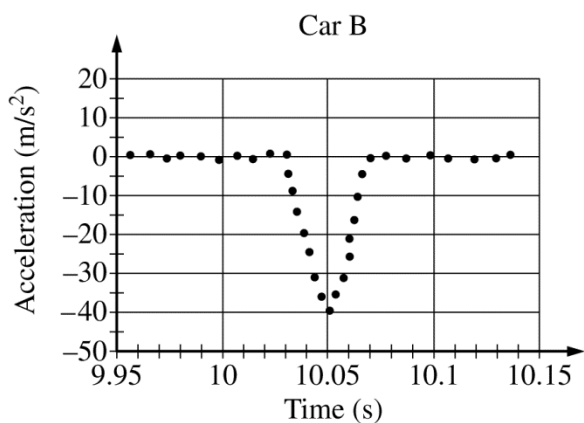
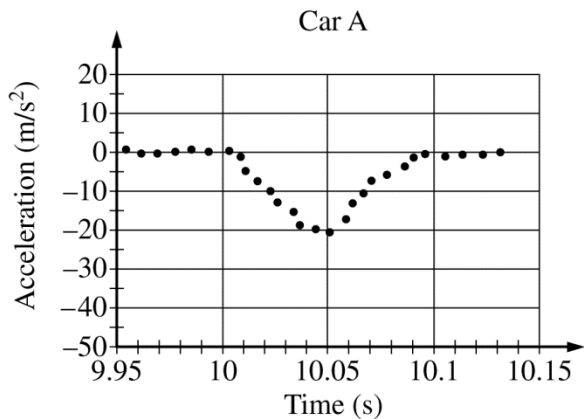


36. The graphs above show the magnitude F of a force exerted on an object as a function of the object's position x for two trials in an experiment. W_1 and W_2 are the work done on the object by force 1 and force 2, respectively. How do W_1 and W_2 compare, and why?
- (A) $W_1 > W_2$, because the maximum value of force 1 is greater than the maximum value of force 2.
- (B) $W_1 > W_2$, because the slope of force 1's graph increases, while the slope of force 2's graph decreases.
- (C) $W_1 < W_2$, because the average value of force 1 is smaller than the average value of force 2.
- (D) $W_1 < W_2$, because at the midpoint, $x = 0.5$ m, the value of force 1 is less than the value of force 2.

Directions: For each of the questions or incomplete statements below, two of the suggested answers will be correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on page 3 of the answer sheet.

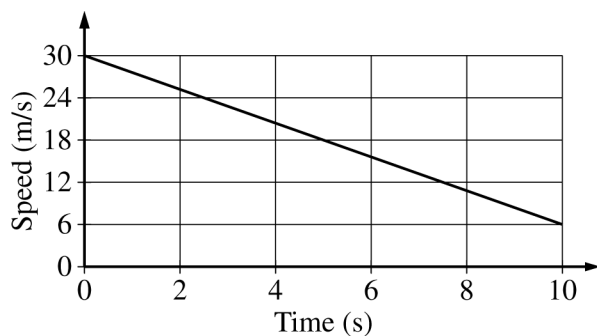


131. A water-skier with weight $F_g = mg$ moves to the right with acceleration a . A horizontal tension force T is exerted on the skier by the rope, and a horizontal drag force F_d is exerted by the water on the ski. The water also exerts a vertical lift force L on the skier. Which of the following are correct relationships between the forces exerted on the skier-ski system? Select two answers.
- (A) $T - F_d = ma$
 - (B) $L - F_g = ma$
 - (C) $L - F_g = 0$
 - (D) $T - F_d = 0$



132. Two model cars, A and B, have the same mass but different bumpers. The acceleration of each car during its collision with a wall is measured, and the data are shown in the graphs above. Which of the following statements about the collisions are correct? Select two answers.

- (A) Both cars reach their maximum speed at 10.05 s.
- (B) The cars experience approximately the same impulse.
- (C) Car B experiences a nonzero force for a longer time than car A.
- (D) The change in momentum for car B occurs over a shorter period of time than for car A.



133. The graph above shows the speed of a truck as it moves along a straight, level road. Which of the following describe a method to determine the distance d the truck travels during the 10 s time interval shown? Select two answers.
- (A) Multiply the average speed of 18 m/s by the 10 s travel time.
 - (B) Multiply half the initial speed of 30 m/s by the 10 s travel time.
 - (C) Calculate the slope of the line in the graph to determine the acceleration a and then use $d = (1/2)at^2$, where t is the travel time.
 - (D) Calculate the area under the line in the graph.

134. A vehicle lands on Mars and explores its surface. The average gravitational field on the surface of Mars is 3.7 N/kg . The weight of the vehicle is defined as the gravitational force exerted on it. Which of the following statements are true about the vehicle's weight? Select two answers.
- (A) The vehicle's weight was constant until it reached the surface of Mars.
 - (B) The vehicle's weight increased while it was descending to the surface of Mars.
 - (C) The vehicle's weight always equals the normal force exerted by Mars on the vehicle while it is landing.
 - (D) The vehicle weighs less on the surface of Mars than on the surface of Earth.

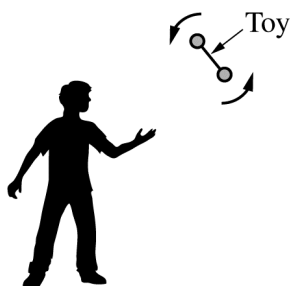
PHYSICS 1

Section II

Time—1 hour and 30 minutes

5 Questions

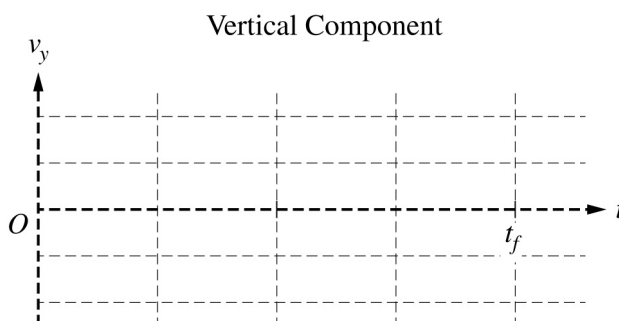
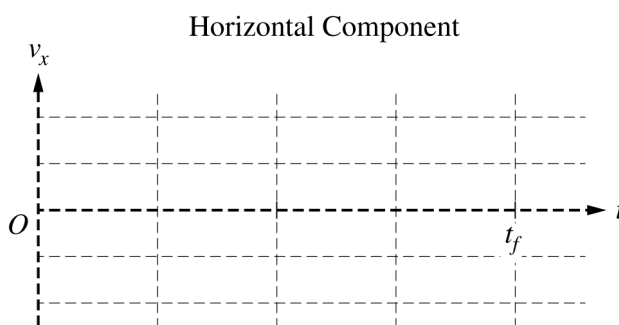
Directions: Questions 1, 4, and 5 are short free-response questions that require about 13 minutes each to answer and are worth 7 points each. Questions 2 and 3 are long free-response questions that require about 25 minutes each to answer and are worth 12 points each. Show your work for each part in the space provided after that part.

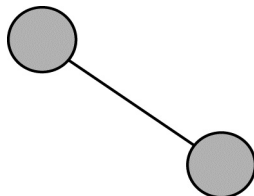


1. (7 points, suggested time 13 minutes)

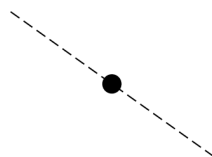
A toy consists of two identical solid spheres connected by a string with negligible mass. The toy is thrown at an angle above the horizontal (not straight up) such that the string remains taut and both spheres are revolving counterclockwise in a vertical plane around the center of the string, as shown above.

- (a) Sketch graphs of the horizontal and vertical components of the velocity of the center of the string as a function of time, from the instant the spheres are released at time $t = 0$ until the instant the system returns to its initial height at time t_f . Take the positive direction to be toward the right for the horizontal component and the positive direction to be upward for the vertical component.



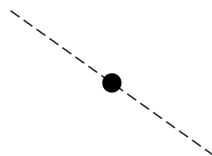


- (b) The figure above shows the toy at the instant the center of the string reaches the top of its trajectory. This is a side view: the sphere on the left is higher than the sphere on the right.
- On the dot below, which represents the left sphere only, draw and label the forces (not components) exerted on the left sphere at this instant. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed line is drawn at the same angle as the string.



Left Sphere

- On the dot below, which represents the whole toy (the spheres-string system), draw and label the forces (not components) that act on the system at this instant. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed line is drawn at the same angle as the string.



Whole Toy

- When the toy was released, the center of the string was moving with an initial speed of 15 m/s at a 60° angle above the horizontal. Calculate the speed of the center of the string at the instant shown above, when the center of the string reaches the top of its trajectory.

2. (12 points, suggested time 25 minutes)

A heavy lab cart moves with kinetic energy K_{init} on a track and collides with a lighter lab cart that is initially at rest. The carts bounce off each other but the collision is not perfectly elastic, causing the two-cart system to lose kinetic energy K_{lost} . A student wonders if the fraction of kinetic energy lost from the two-cart system during the collision $\left(\frac{K_{\text{lost}}}{K_{\text{init}}}\right)$ depends on the speed of the first cart before the collision and plans to perform an experiment.

- (a) The student hypothesizes that a greater fraction of kinetic energy is lost from the system during the collision when the speed of the first cart is greater.

Briefly state one reason the hypothesis might be correct.

- (b) Design an experimental procedure that could be used to test the student's hypothesis. Assume equipment usually found in a school physics laboratory is available.

In the table below, list the quantities that would be measured and the equipment that would be used to measure each quantity. Also, define a symbol to represent each quantity. You do not need to use every row and may add additional rows as needed.

Quantity to be Measured	Symbol	Equipment for Measurement

Describe the overall procedure to be used, referring to the table above. Provide enough detail so that another student could replicate the experiment. As needed, use the symbols defined in the table and/or include a simple diagram of the setup. Be sure to address how experimental uncertainty could be reduced.

- (c) Describe how the experimental data could be analyzed to confirm or disconfirm the hypothesis that a greater fraction of kinetic energy is lost from the system during the collision when the speed of the first cart is greater. Include a description or example of any equations, data tables, graphs, or other representations that could be used.

- (d) Consider a different scenario in which the carts stick together after the collision. The masses of the heavier and lighter cart are m_1 and m_2 , respectively. Derive an expression for the fraction of kinetic energy lost $\left(\frac{K_{\text{lost}}}{K_{\text{init}}}\right)$ during the collision. Express your answer in terms of m_1 and m_2 .



Note: Figure not drawn to scale.

3. (12 points, suggested time 25 minutes)

As shown above, the left end of a horizontal string is attached to an oscillator and the right end is attached to a wall. The string's midpoint is labeled C , and a dot has been painted on the left half of the string. The left half of the string is uniformly thick, with diameter D_0 , while the right half of the string gradually increases in thickness from the midpoint C to the wall. The string is made of a material with uniform density (mass per unit volume).

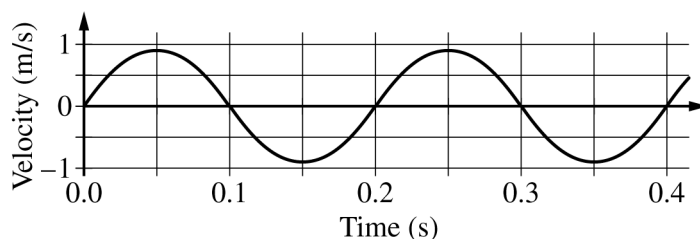
(a) The oscillator shakes the left end of the string up and down, creating waves traveling to the right with speed v_0 . A student observes that every wave crest approaching the midpoint C passes through C but gradually slows down on the right half of the string.

i. On which half of the string, if either, do the waves have a longer wavelength?

____ Left half ____ Right half ____ Same wavelength on both halves

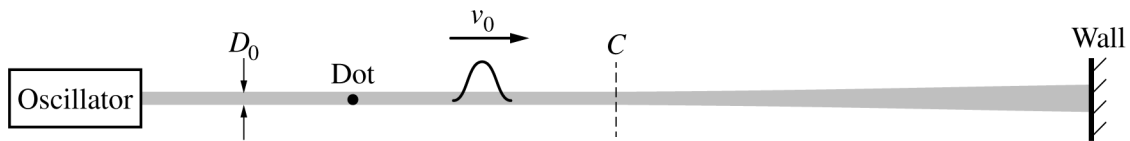
Briefly explain your reasoning.

ii. The graph below represents the vertical velocity of the dot on the left half of the string as a function of time.



Describe how you could use the graph to estimate the amplitude of the waves at the dot's location. You do NOT have to actually calculate the amplitude.

(b) In a different experiment, the oscillator shakes the left end of the string up and down at a rate of 50 cycles per second. The leading edge of the wave reaches the midpoint C , which is 0.75 m from the left end of the string, 0.030 second after the oscillator starts shaking the string. Calculate the wavelength of the wave on the left half of the string.



Note: Figure not drawn to scale.

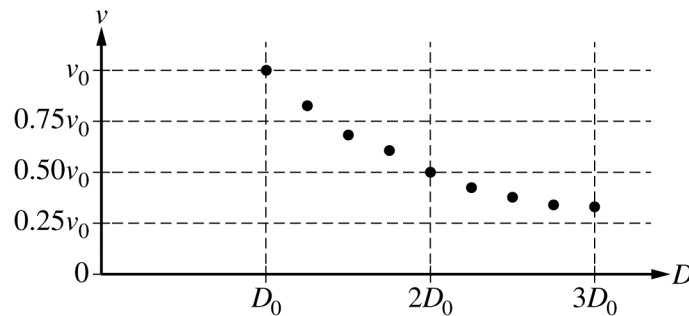
Later, the student uses the oscillator to create a single pulse with initial speed v_0 at the left end of the string, as shown above. The student wants to quantify the observation that the pulse slows down on the right half of the string. The student comes up with the following equation relating the string's diameter D at a given point on the string to the speed v of the pulse at that point: $v = v_0 \left(\frac{D_0}{D} \right)$.

(c) Is the equation consistent with the observation that the pulse slows down on the right half of the string?

☐ Yes ☐ No

Briefly explain your reasoning.

(d) Using frame-by-frame analysis of video of the pulse, the student plots the speed v of the pulse as a function of the string diameter D , as shown below.



Are the data in the graph consistent with the student's equation above?

☐ Yes ☐ No

Briefly explain your answer.

Question 3 continues on the next page.

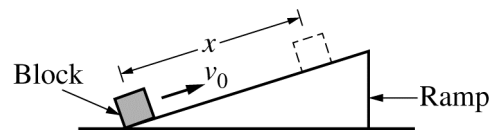


Note: Figure not drawn to scale.

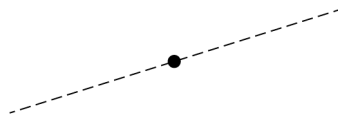
- (e) The figure above shows only the right half of the string. The solid line represents the shape of the wave pulse when its peak reaches point X , which is to the right of the midpoint C . Draw the pulse when it reaches point Y . Your drawing should clearly indicate whether the width of the pulse at point Y is greater than, equal to, or less than the width of the pulse at point X .

4. (7 points, suggested time 13 minutes)

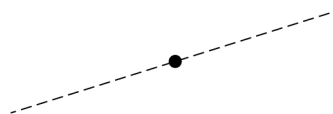
A student strikes a block at the bottom of a ramp, giving it an initial speed v_0 up the ramp, as shown at right. There is friction between the ramp and the block as it slides a distance x up the ramp and then slides back down.



- (a) On the dots below, which represent the block as it is sliding up the ramp and down the ramp, draw and label the forces (not components) exerted on the block. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed lines are drawn at the same angle as the surface of the ramp.



Up the Ramp



Down the Ramp

- (b) The block takes time t_{up} to slide up the ramp a distance x . The block then takes time t_{down} to slide back down to the bottom of the ramp, where it has speed v_f . Is t_{down} greater than, equal to, or less than t_{up} ?

_____ $t_{\text{down}} > t_{\text{up}}$ _____ $t_{\text{down}} = t_{\text{up}}$ _____ $t_{\text{down}} < t_{\text{up}}$

In a clear, coherent paragraph-length response that may also contain figures and/or equations, explain your reasoning. If you need to draw anything other than what you have shown in part (a) to assist in your response, use the space below. Do NOT add anything to the figures in part (a).

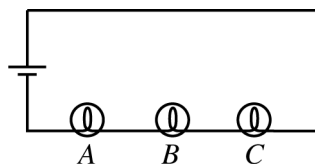


Figure 1

5. (7 points, suggested time 13 minutes)

Students in a physics lab use a battery and three identical lightbulbs to construct the circuit shown in Figure 1 above. The brightness of each bulb increases when the current through the bulb increases.

(a) Which bulb, if any, is the brightest?

___ A ___ B ___ C ___ All bulbs have the same brightness.

Briefly explain your reasoning in terms of current and/or potential difference.

(b) The students observe the brightness of bulbs *A*, *B*, and *C*. The students then add a fourth lightbulb *D*, with a larger resistance than bulbs *A*, *B*, and *C*, as shown in Figure 2 below.

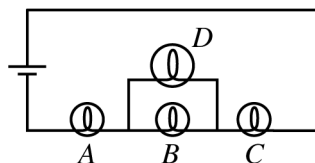


Figure 2

i. Is bulb *A* now brighter than, dimmer than, or the same brightness as it was in the circuit in Figure 1 ?

___ Brighter ___ Dimmer ___ Same brightness

Briefly explain your reasoning.

ii. Is bulb *B* now brighter than, dimmer than, or the same brightness as it was in the circuit in Figure 1 ?

___ Brighter ___ Dimmer ___ Same brightness

Briefly explain your reasoning.

- (c) Bulbs A , B , and C each have resistance $10\ \Omega$, while bulb D has a resistance of $30\ \Omega$. In the circuit shown in Figure 2, the current through bulb D is measured to be $0.20\ \text{A}$. Calculate the potential difference across bulb A .

Answer Key for AP Physics 1
Practice Exam, Section I

Question 1: C	Question 21: A
Question 2: B	Question 22: C
Question 3: B	Question 23: C
Question 4: D	Question 24: A
Question 5: D	Question 25: D
Question 6: C	Question 26: A
Question 7: A	Question 27: C
Question 8: C	Question 28: D
Question 9: B	Question 29: A
Question 10: C	Question 30: C
Question 11: B	Question 31: B
Question 12: C	Question 32: C
Question 13: C	Question 33: B
Question 14: D	Question 34: D
Question 15: C	Question 35: B
Question 16: D	Question 36: C
Question 17: C	Question 131: A,C
Question 18: A	Question 132: B,D
Question 19: D	Question 133: A,D
Question 20: A	Question 134: B,D

2018 AP Physics 1: Algebra-Based Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
1	3.A.3.1; 3.B.2.1	3.A.3; 3.B.2	6.4; 1.4	C	89
2	5.D.2.5	5.D.2	2.2	B	52
3	3.A.1.1; 3.A.1.3	3.A.1; 3.A.1	1.5; 2.2; 5.1	B	90
4	3.A.1.1; 3.A.1.3; 3.B.3.3	3.A.1; 3.A.1; 3.B.3	1.5; 2.2; 5.1; 2.2; 5.1	D	67
5	3.B.3.3	3.B.3	2.2; 5.1	D	54
6	6.B.2.1; 6.B.4.1	6.B.2; 6.B.4	1.4; 5.1	C	62
7	6.D.3.1	6.D.3	4.2	A	47
8	3.B.2.1; 4.C.1.1; 4.C.2.2; 5.B.4.2; 5.B.5.5	3.B.2; 4.C.1; 4.C.2; 5.B.4; 5.B.5	2.2; 2.2; 2.2; 2.2; 2.2; 6.4	C	68
9	4.B.2.2	4.B.2	5.1	B	36
10	5.B.9.3; 5.C.3.1; 5.C.3.3	5.B.9; 5.C.3; 5.C.3	6.4; 7.2; 6.4; 7.2; 1.4	C	64
11	3.E.1.1; 3.E.1.2; 5.B.3.1; 5.B.4.1	3.E.1; 3.E.1; 5.B.3; 5.B.4	6.4; 7.2; 1.4; 2.2; 6.4; 7.2; 6.4; 7.2	B	51
12	5.B.4.1; 5.B.4.2; 5.B.5.4	5.B.4; 5.B.4; 5.B.5	6.4; 1.4; 6.4	C	83
13	3.E.1.1; 4.C.1.2	3.E.1; 4.C.1	6.4; 6.4	C	48
14	5.A.2.1; 5.B.4.2	5.A.2; 5.B.4	6.4; 1.4; 2.1; 2.2	D	37
15	1.B.1.1; 1.B.1.2; 5.A.2.1	1.B.1; 1.B.1; 5.A.2	6.4; 6.4; 6.4	C	63
16	3.A.1.1; 3.B.1.3; 3.B.2.1	3.A.1; 3.B.1; 3.B.2	1.5; 2.2; 1.5; 2.2; 1.1; 1.4; 2.2	D	43
17	5.B.4.2	5.B.4	1.4; 2.1	C	49
18	3.E.1.1; 4.C.2.1; 5.B.5.4	3.E.1; 4.C.2; 5.B.5	6.4; 7.2; 6.4; 6.4; 7.2	A	44
19	4.C.1.1; 4.C.1.2; 5.B.3.2; 5.B.5.5	4.C.1; 4.C.1; 5.B.3; 5.B.5	1.4; 2.1; 2.2; 6.4; 1.4; 2.2; 2.2; 6.4	D	32
20	3.A.1.1	3.A.1	1.5; 2.2	A	37
21	3.E.1.1; 3.E.1.4; 4.C.1.1; 4.C.2.2; 5.B.4.2	3.E.1; 3.E.1; 4.C.1; 4.C.2; 5.B.4	6.4; 2.2; 1.4; 2.2; 1.4; 2.2; 1.4; 2.2	A	36
22	3.A.1.1; 5.B.5.5	3.A.1; 5.B.5	2.2; 2.2	C	39
23	3.B.1.1; 3.B.2.1; 4.A.2.1	3.B.1; 3.B.2; 4.A.2	6.4; 1.1; 1.4; 6.4	C	41
24	6.A.1.1	6.A.1	6.2	A	47
25	1.B.1.1; 5.B.9.3; 5.C.3.3	1.B.1; 5.B.9; 5.C.3	6.4; 2.2; 6.4; 7.2; 1.4; 2.2	D	47
26	5.C.3.1; 5.C.3.3	5.C.3; 5.C.3	6.4; 1.4; 2.2	A	36
27	3.F.1.2; 4.D.1.1	3.F.1; 4.D.1	1.4; 1.4	C	58
28	3.F.2.1; 4.D.3.1	3.F.2; 4.D.3	6.4; 2.2	D	43
29	3.F.2.1	3.F.2	6.4	A	32
30	3.B.3.1	3.B.3	6.4; 7.2	C	32
31	6.A.3.1; 6.B.1.1	6.A.3; 6.B.1	1.4; 1.4	B	69
32	6.D.4.2	6.D.4	2.2	C	29
33	3.A.1.1; 3.A.3.1	3.A.1; 3.A.3	1.5; 2.2; 6.4; 7.2	B	33
34	4.A.1.1; 4.A.2.1; 4.A.3.2; 5.D.2.5; 5.D.3.1	4.A.1; 4.A.2; 4.A.3; 5.D.2; 5.D.3	1.4; 6.4; 6.4; 1.4; 2.2; 6.4	D	25
35	3.C.2.1	3.C.2	6.4	B	31
36	5.B.5.3	5.B.5	1.4; 2.2; 6.4	C	42

2018 AP Physics 1: Algebra-Based Question Descriptors and Performance Data

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
131	3.A.4.3; 3.B.2.1	3.A.4; 3.B.2	1.4; 1.1; 1.4	A, C	66
132	3.D.2.2; 3.D.2.3	3.D.2; 3.D.2	6.4; 5.1	B, D	40
133	3.A.1.1	3.A.1	1.5; 2.2	A, D	36
134	2.B.1.1; 2.B.2.1	2.B.1; 2.B.2	2.2; 7.2; 2.2	B, D	42

Free-Response Questions

Question	Learning objectives	Essential Knowledge	Science Practices	Mean Score
1	3.A.4.3; 3.B.2.1; 4.A.2.1; 4.A.2.3; 4.A.3.2	3.A.4; 3.B.2; 4.A.2; 4.A.2; 4.A.3	1.4; 1.1; 6.4; 1.4; 2.2; 1.4	3.61
2	3.A.1.2; 4.C.1.1; 5.B.4.2; 5.D.1.4; 5.D.2.5	3.A.1; 4.C.1; 5.B.4; 5.D.1; 5.D.2	4.2; 1.4; 2.2; 1.4; 2.2; 4.2; 5.1; 6.4; 2.2	3.06
3	3.A.1.1; 3.B.3.1; 3.B.3.4; 6.A.1.2; 6.B.4.1	3.A.1; 3.B.3; 3.B.3; 6.A.1; 6.B.4	1.5; 2.1; 2.2; 6.4; 7.2; 2.2; 6.2; 1.2; 5.1; 7.2	5.24
4	2.B.1.1; 3.A.1.1; 3.A.2.1; 3.B.1.1; 3.B.1.3; 3.B.2.1	2.B.1; 3.A.1; 3.A.2; 3.B.1; 3.B.1; 3.B.2	2.2; 1.5; 2.2; 1.1; 6.4; 7.2; 1.5; 2.2; 1.1; 1.4; 2.2	1.62
5	5.B.9.3; 5.C.3.1; 5.C.3.3	5.B.9; 5.C.3; 5.C.3	2.2; 6.4; 6.4; 1.4; 2.2	2.65

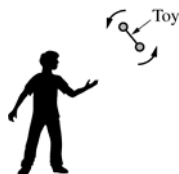
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Question 1

7 points total

**Distribution
of points**



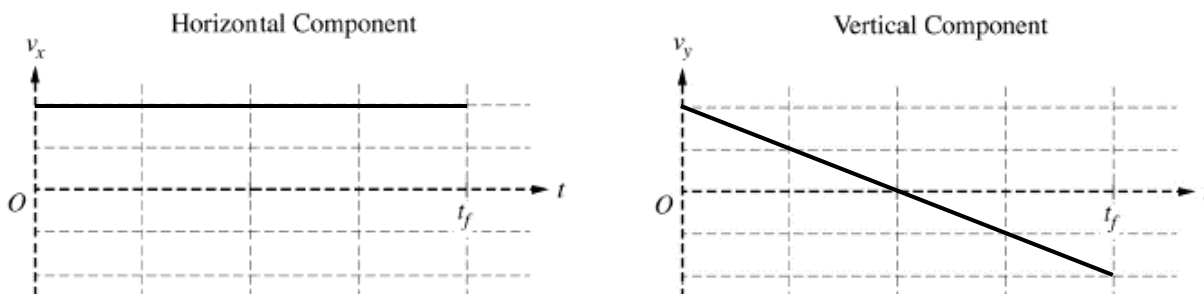
A toy consists of two identical solid spheres connected by a string with negligible mass. The toy is thrown at an angle above the horizontal (not straight up) such that the string remains taut and both spheres are revolving counterclockwise in a vertical plane around the center of the string, as shown above.

(a) LO / SP: 4.A.2.1 / 6.4; 4.A.2.3 / 1.4, 2.2

2 points

Sketch graphs of the horizontal and vertical components of the velocity of the center of the string as a function of time, from the instant the spheres are released at time $t = 0$ until the instant the system returns to its initial height at time t_f . Take the positive direction to be toward the right for the horizontal component and the positive direction to be upward for the vertical component.

Example graphs:



Horizontal component: For a non-zero horizontal line		1 point
Vertical component: For a non-vertical line with a negative slope that crosses the horizontal axis near $t_f/2$ and reaches t_f		1 point
<u>Note:</u> Any part of the sketched graph beyond t_f is ignored.		

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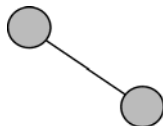
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Question 1 (continued)

**Distribution
of points**

(b) LO / SP: 3.A.4.3 / 1.4; 3.B.2.1 / 1.1; 4.A.2.1 / 6.4; 4.A.2.3 / 1.4, 2.2; 4.A.3.2 / 1.4

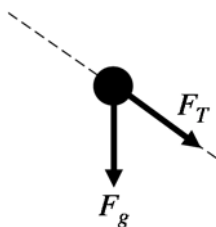
5 points



The figure above shows the toy at the instant the center of the string reaches the top of its trajectory. This is a side view: the sphere on the left is higher than the sphere on the right.

i 2 points

On the dot below, which represents the left sphere only, draw and label the forces (not components) exerted on the left sphere at this instant. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed line is drawn at the same angle as the string.

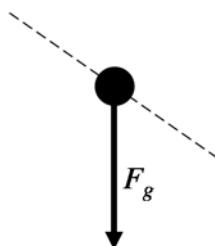


Left Sphere

For a gravitational force with correct direction and label		1 point
For a tension force with correct direction and label		1 point
<u>Note:</u> A maximum of one point may be earned if there are any extraneous forces.		

ii 1 point

On the dot below, which represents the whole toy (the spheres-string system), draw and label the forces (not components) that act on the system at this instant. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed line is drawn at the same angle as the string.



Whole Toy

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Question 1 (continued)

**Distribution
of points**

(b)(ii) continued

For a vector pointing downward representing the gravitational force with no extraneous forces		1 point
<u>Note:</u> A label is not required, since the use of labels is assessed in part (b)(i).		

iii 2 points

When the toy was released, the center of the string was moving with an initial speed of 15 m/s at a 60° angle above the horizontal. Calculate the speed of the center of the string at the instant shown above, when the center of the string reaches the top of its trajectory.

For attempting to find the horizontal component of velocity <u>Note:</u> See examples below. This point is earned for the wrong use of trig functions if it is clear it is being used to find the horizontal component, but not awarded if it's not clear which component the trig function is referring to, or for stating that the vertical component is zero.		1 point
For a correct numerical answer for horizontal velocity with correct units		1 point
Examples: 0 points: $v \sin \theta = 13 \text{ m/s}$ (No clear indication of an attempt to find the horizontal component) 1 point: $v_x = v \sin \theta = 13 \text{ m/s}$ (An attempt to find the horizontal component is implied by the v_x term) 1 point: $v \cos \theta$ (An attempt to find the horizontal component is implied by the $\cos \theta$ term) 1 point: $v \cos \theta = 7.5$ (Correct numerical answer with no units) 1 point: 7.5 m/s (Correct answer with units, but no work shown) 2 points: $15 \cos 60 = 7.5 \text{ m/s}$ (Correct answer with units) 2 points: $15 \cos 60 \text{ m/s}$ (Correct answer with units)		

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Question 1 (continued)

Learning Objectives (LO)

LO 3.A.4.3: The student is able to analyze situations involving interactions among several objects by using free-body diagrams that include the application of Newton’s third law to identify forces. [See Science Practice 1.4]

LO 3.B.2.1: The student is able to create and use free-body diagrams to analyze physical situations to solve problems with motion qualitatively and quantitatively. [See Science Practices 1.1, 1.4, 2.2]

LO 4.A.2.1: The student is able to make predictions about the motion of a system based on the fact that acceleration is equal to the change in velocity per unit time, and velocity is equal to the change in position per unit time. [See Science Practice 6.4]

LO 4.A.2.3: The student is able to create mathematical models and analyze graphical relationships for acceleration, velocity, and position of the center of mass of a system and use them to calculate properties of the motion of the center of mass of a system. [See Science Practices 1.4, 2.2]

LO 4.A.3.2: The student is able to use visual or mathematical representations of the forces between objects in a system to predict whether or not there will be a change in the center-of-mass velocity of that system. [See Science Practice 1.4]

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Question 2

12 points total

**Distribution
of points**

A heavy lab cart moves with kinetic energy K_{init} on a track and collides with a lighter lab cart that is initially at rest. The carts bounce off each other but the collision is not perfectly elastic, causing the two-cart system to lose kinetic energy K_{lost} . A student wonders if the fraction of kinetic energy lost from the two-cart system during the collision $\left(\frac{K_{\text{lost}}}{K_{\text{init}}}\right)$ depends on the speed of the first cart before the collision and plans to perform an experiment.

(a) LO / SP: 5.D.1.4 / 4.2, 6.4
1 point

The student hypothesizes that a greater fraction of kinetic energy is lost from the system during the collision when the speed of the first cart is greater.

Briefly state one reason the hypothesis might be correct.

For a valid statement relating the speed of cart and dissipation of mechanical energy <u>Note for pre-reading:</u> Response does not have to address the fractional loss of kinetic energy (points are earned for this distinction in later parts).		1 point
Examples of valid statements: More damage when going faster Slower collision means compression and decompression during collision happen without permanent deformation Examples of invalid statements: Higher fraction of K is lost at higher speeds Because more kinetic energy is lost		

(b) LO / SP: 3.A.1.2 / 4.2; 5.D.1.4 / 4.2
5 points

Design an experimental procedure that could be used to test the student's hypothesis. Assume equipment usually found in a school physics laboratory is available.

In the table below, list the quantities that would be measured and the equipment that would be used to measure each quantity. Also, define a symbol to represent each quantity. You do not need to use every row and may add additional rows as needed.

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Question 2 (continued)

**Distribution
of points**

Quantity to be Measured	Symbol	Equipment for Measurement

Describe the overall procedure to be used, referring to the table above. Provide enough detail so that another student could replicate the experiment. As needed, use the symbols defined in the table and/or include a simple diagram of the setup. Be sure to address how experimental uncertainty could be reduced.

For measuring the speed of both carts before and after collision <u>Note:</u> It is not necessary to measure the initial speed of the cart that is at rest.	1 point
For measuring the mass of the carts (or calculating mass ratio using conservation of momentum, which may be stated later)	1 point
For equipment and measurements consistent with procedure as described or drawn in diagram (e.g., motion detectors in the right places to measure speed of both carts)	1 point
For varying the speed (making speed the independent variable) of the first cart in a feasible experiment that could be done in a school lab. <u>Note:</u> “Varying the speed” means at least 2 trials (to test hypothesis), and “feasible” means that equipment is used appropriately for each measurement.	1 point
For attempting to reduce uncertainty, e.g., multiple trials for a given initial speed of cart 1, or trials involving at least three different initial speeds	1 point

Example:

Quantity to be Measured	Symbol	Equipment for Measurement
Initial speed of heavy cart (H)	v_{iH}	Motion sensor (MS1)
Final speed of heavy cart	v_{fH}	Motion sensor (MS1)
Mass of heavy cart	m_H	Triple beam balance
Final speed of light cart (L)	v_{fL}	Motion sensor (MS2)
Mass of light cart	m_L	Triple beam balance



Measure the mass of each cart with the triple beam balance. Set the carts and motion sensors on the track. With the motion sensors recording, push the heavy cart toward the light cart so that they collide. Record the motion sensors readings for the speeds of the heavy cart immediately before the collision, and the speeds of both carts immediately after the collision. Repeat 8 times, varying the force with which the heavy cart is pushed so that a wide range of initial velocities for the heavy cart are used.

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Question 2 (continued)

**Distribution
of points**

(c) LO / SP: 4.C.1.1 / 1.4, 2.2; 5.B.4.2 / 1.4, 2.2; 5.D.1.4 / 5.1

3 points

Describe how the experimental data could be analyzed to confirm or disconfirm the hypothesis that a greater fraction of kinetic energy is lost from the system during the collision when the speed of the first cart is greater. Include a description or example of any equations, data tables, graphs, or other representations that could be used.

For a correct indication of how initial and final total kinetic energy would be calculated from the raw data	1 point
For using the difference between initial and final total kinetic energy to determine the kinetic energy lost	1 point
For a description of how the calculated <u>fractional</u> energy losses would be used to confirm or disconfirm the hypothesis	1 point
Example: For each trial, calculate the initial kinetic energy of the heavy cart using $K_{\text{init}} = \frac{1}{2} m_{\text{H}} v_{\text{iH}}^2$. Also calculate the final kinetic energy $K_{\text{f}} = \frac{1}{2} m_{\text{H}} v_{\text{fH}}^2 + \frac{1}{2} m_{\text{L}} v_{\text{fL}}^2$. Make a graph of $K_{\text{lost}}/K_{\text{init}} = (K_{\text{init}} - K_{\text{f}})/K_{\text{init}}$ as a function of v_{iH}. If a positive trend is seen in the data, the hypothesis is confirmed.	

(d) LO / SP: 4.C.1.1 / 1.4, 2.2; 5.B.4.2 / 1.4, 2.2; 5.D.2.5 / 2.2

3 points

Consider a different scenario in which the carts stick together after the collision. The masses of the heavier and lighter cart are m_1 and m_2 , respectively. Derive an expression for the fraction of kinetic energy lost $\left(\frac{K_{\text{lost}}}{K_{\text{init}}}\right)$ during the collision. Express your answer in terms of m_1 and m_2 .

For using momentum conservation to find v_{f} in terms of v_{i} , or the ratio of post-collision to pre-collision speed	1 point
For a correct expression for final kinetic energy (or consistent with v_{f} found from conservation of momentum) OR a correct expression for the fractional energy loss in terms of m_1 , m_2 , the initial speed of the heavy cart, and the final speed of the two-cart system	1 point
A correct expression for fraction of kinetic energy lost: $\frac{m_2}{m_1 + m_2} \text{ or } 1 - \frac{m_1}{m_1 + m_2}$ <u>Note:</u> The above point is not earned if the answer includes any variables other than m_1 and m_2	1 point

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Question 2 (continued)

Learning Objectives (LO)

LO 3.A.1.2: The student is able to design an experimental investigation of the motion of an object. [See Science Practice 4.2]

LO 4.C.1.1: The student is able to calculate the total energy of a system and justify the mathematical routines used in the calculation of component types of energy within the system whose sum is the total energy. [See Science Practices 1.4, 2.1, and 2.2]

LO 5.B.4.2: The student is able to calculate changes in kinetic energy and potential energy of a system, using information from representations of that system. [See Science Practices 1.4, 2.1, and 2.2]

LO 5.D.1.4: The student is able to design an experimental test of an application of the principle of the conservation of linear momentum, predict an outcome of the experiment using the principle, analyze data generated by that experiment whose uncertainties are expressed numerically, and evaluate the match between the prediction and the outcome. [See Science Practices 4.2, 5.1, 5.3, and 6.4]

LO 5.D.2.5: The student is able to classify a given collision situation as elastic or inelastic, justify the selection of conservation of linear momentum as the appropriate solution method for an inelastic collision, recognize that there is a common final velocity for the colliding objects in the totally inelastic case, solve for missing variables, and calculate their values. [See Science Practices 2.1 and 2.2]

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Question 3

12 points total

**Distribution
of points**



Note: Figure not drawn to scale.

As shown above, the left end of a horizontal string is attached to an oscillator and the right end is attached to a wall. The string's midpoint is labeled C , and a dot has been painted on the left half of the string. The left half of the string is uniformly thick, with diameter D_0 , while the right half of the string gradually increases in thickness from the midpoint C to the wall. The string is made of a material with uniform density (mass per unit volume).

(a) LO / SP: 3.A.1.1 / 1.5, 2.1, 2.2; 3.B.3.1 / 6.4, 7.2; 3.B.3.4 / 2.2, 6.2; 6.B.4.1 / 5.1, 7.2

5 points

The oscillator shakes the left end of the string up and down, creating waves traveling to the right with speed v_0 . A student observes that every wave crest approaching the midpoint C passes through C but gradually slows down on the right half of the string.

i 2 points

On which half of the string, if either, do the waves have a longer wavelength?

____ Left half ____ Right half ____ Same wavelength on both halves

Briefly explain your reasoning.

Correct answer: "Left half"		
<u>Note:</u> A maximum of 1 point may be earned if the wrong selection is made.		
For stating that the frequency stays the same		1 point
For stating that, since the speed of the wave decreases as it moves to the right, the wavelength must decrease as well, <u>or</u> for using $\lambda \propto v$ or $v = f\lambda$		1 point

<i>Alternate solution</i>		
<i>For using reasoning that applies to a pulse (the leading edge is slower than trailing edge, so the trailing edge "catches up to" the leading edge and wavelength decreases)</i>		2 points

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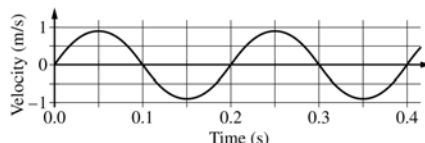
Question 3 (continued)

**Distribution
of points**

(a) continued

ii 3 points

The graph below represents the vertical velocity of the dot on the left half of the string as a function of time.



Describe how you could use the graph to estimate the amplitude of the waves at the dot's location. You do NOT have to actually calculate the amplitude.

For using the area under the velocity vs. time curve.		1 point
For correctly or incorrectly relating the area under the curve to the amplitude OR identifying a specific area (in terms of range of times, “humps”, cycles, etc.)		1 point
For correctly identifying a quarter cycle, or “half a hump”, as the area or displacement corresponding to the amplitude (or any correct relationship between cycles and amplitude)		1 point
Examples: 1 point: Unspecified area: “Use the area under the curve.” 2 points: Any area specified: “Use the area under the curve from $t = 0$ to $t = 0.4$ s” “Area under a full hump because that’s a full period.” “Area under the curve for 1/4 period” (only 2 points because didn’t specify what period corresponds to on the graph) 3 points: Correct area (quarter cycle, OR half cycle with statement that it’s twice the amplitude): “The area under half a hump”		

<i>Alternate solution based on $v_{\max} = \omega A$</i>		
<i>For using $v_{\max} = \omega A$</i>		1 point
<i>For indicating how ω can be determined from the graph (e.g., $\omega = \frac{2\pi}{T}$, where T corresponds to a complete cycle)</i>		1 point
<i>For a correct algebraic expression for amplitude (e.g., $\frac{v_{\max} T}{2\pi}$)</i>		1 point

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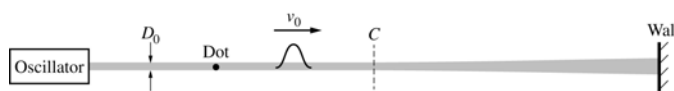
Question 3 (continued)

**Distribution
of points**

(b) LO / SP: 3.A.1.1 / 1.5, 2.1, 2.2; 6.B.4.1 / 5.1
3 points

In a different experiment, the oscillator shakes the left end of the string up and down at a rate of 50 cycles per second. The leading edge of the wave reaches the midpoint C , which is 0.75 m from the left end of the string, 0.030 second after the oscillator starts shaking the string. Calculate the wavelength of the wave on the left half of the string.

For calculating wave speed ($v = d/t$)		1 point
For implicitly or explicitly using $f = 50$ Hz		1 point
For calculating wavelength using $\lambda = v/f$		1 point
Example: $v = \frac{0.75 \text{ m}}{0.030 \text{ s}} = 25 \text{ m/s}$ $\lambda = \frac{v}{f} = \frac{25 \text{ m/s}}{50 \text{ s}^{-1}} = 0.5 \text{ m}$		



Note: Figure not drawn to scale.

Later, the student uses the oscillator to create a single pulse with initial speed v_0 at the left end of the string, as shown above. The student wants to quantify the observation that the pulse slows down on the right half of the string. The student comes up with the following equation relating the string's diameter D at a given point on the string to the speed v of the pulse at that point: $v = v_0 \left(\frac{D_0}{D} \right)$.

(c) LO / SP: 3.A.1.1 / 1.5, 2.2
1 point

Is the equation consistent with the observation that the pulse slows down on the right half of the string?

☐ Yes ☐ No

Briefly explain your reasoning.

Correct answer: "Yes"

Note: If the wrong selection is made, the explanation is not graded.

For stating that as D increases, v decreases		1 point
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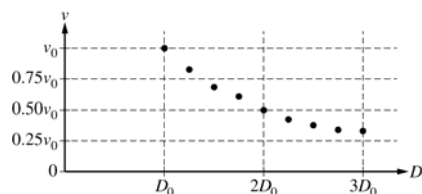
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Question 3 (continued)

**Distribution
of points**

(d) LO / SP: 3.A.1.1 / 1.5, 2.2
1 point

Using frame-by-frame analysis of video of the pulse, the student plots the speed v of the pulse as a function of the string diameter D , as shown below.



Are the data in the graph consistent with the student's equation above?

☐ Yes ☐ No

Briefly explain your answer.

Correct answer: "Yes"		
<u>Note:</u> If the wrong selection is made, no credit can be earned.		
For stating that the data agree with the equation because of the downward trend in the data, or for stating that v decreases as D increases (inverse relationship), or for comparing calculated values from the equation with the data in the graph		1 point

(e) LO / SP: 6.A.1.2 / 1.2
2 points



Note: Figure not drawn to scale.

The figure above shows only the right half of the string. The solid line represents the shape of the wave pulse when its peak reaches point X , which is to the right of the midpoint C . Draw the pulse when it reaches point Y . Your drawing should clearly indicate whether the width of the pulse at point Y is greater than, equal to, or less than the width of the pulse at point X .

Example:



Note: Figure not drawn to scale.

For a pulse with similar overall shape as the original		1 point
For a pulse that is narrower <u>Note:</u> The height is not considered for earning this point.		1 point

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Question 3 (continued)

Learning Objectives (LO)

LO 3.A.1.1: The student is able to express the motion of an object using narrative, mathematical, and graphical representations. [See Science Practices 1.5, 2.1, and 2.2]

LO 3.B.3.1: The student is able to predict which properties determine the motion of a simple harmonic oscillator and what the dependence of the motion is on those properties. [See Science Practices 6.4 and 7.2]

LO 3.B.3.4: The student is able to construct a qualitative and/or a quantitative explanation of oscillatory behavior given evidence of a restoring force. [See Science Practices 2.2 and 6.2]

LO 6.A.1.2: The student is able to describe representations of transverse and longitudinal waves. [See Science Practice 1.2]

LO 6.B.4.1: The student is able to design an experiment to determine the relationship between periodic wave speed, wavelength, and frequency and relate these concepts to everyday examples. [See Science Practices 4.2, 5.1, and 7.2]

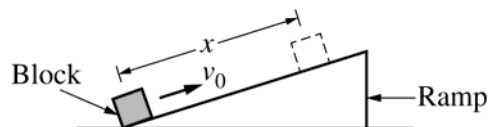
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Question 4

7 points total

**Distribution
of points**

A student strikes a block at the bottom of a ramp, giving it an initial speed v_0 up the ramp, as shown at right. There is friction between the ramp and the block as it slides a distance x up the ramp and then slides back down.

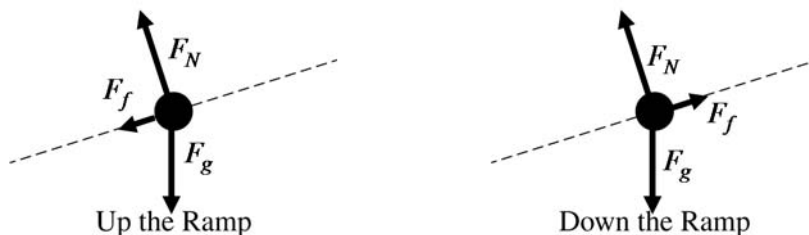


(a) LO / SP: 3.A.2.1 / 1.1; 3.B.2.1 / 1.1, 1.4

2 points

On the dots below, which represent the block as it is sliding up the ramp and down the ramp, draw and label the forces (not components) exerted on the block. Represent each force by a distinct arrow starting on, and pointing away from, the dot. The dashed lines are drawn at the same angle as the surface of the ramp.

Example:



For a labeled friction force vector in the correct direction in both diagrams	1 point
For labeled gravitational and normal forces in the correct directions in both diagrams, with no extraneous forces	1 point

(b) LO / SP: 2.B.1.1 / 2.2; 3.A.1.1 / 1.5, 2.2; 3.B.1.1 / 6.4, 7.2; 3.B.1.3 / 1.5, 2.2; 3.B.2.1 / 1.1, 1.4, 2.2

5 points

The block takes time t_{up} to slide up the ramp a distance x . The block then takes time t_{down} to slide back down to the bottom of the ramp, where it has speed v_f . Is t_{down} greater than, equal to, or less than t_{up} ?

_____ $t_{\text{down}} > t_{\text{up}}$ _____ $t_{\text{down}} = t_{\text{up}}$ _____ $t_{\text{down}} < t_{\text{up}}$

In a clear, coherent paragraph-length response that may also contain figures and/or equations, explain your reasoning. If you need to draw anything other than what you have shown in part (a) to assist in your response, use the space below. Do NOT add anything to the figures in part (a).

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Question 4 (continued)

**Distribution
of points**

Correct Answer: $t_{\text{down}} > t_{\text{up}}$

Note: The response is graded even if an incorrect selection is made.

For stating that the magnitude of the net force on the block is greater when it is sliding up the ramp than when it is sliding down the ramp because the direction of the frictional force changes while the direction of the component of the gravitational force along the ramp does not (this can be implied) OR a description of the net force consistent with the free-body diagrams (FBDs) in part (a)		1 point
For stating that the magnitude of acceleration of the block while sliding up the ramp is greater than that when sliding down, OR a description of acceleration consistent with the FBDs in part (a)		1 point
For a justification that v_f is less than v_0 , OR average v up is greater than average v down (e.g., speed changes more on way up than on way down because acceleration is greater on the way up and the same distance covered and final/initial speed on way up/down is zero), OR a description of final and initial speeds consistent with the FBDs in part (a)		1 point
For a correct <i>argument</i> that, if v_f is less than v_0 or the average speed up is greater than the average speed down, then t_{down} is greater than t_{up} . (This argument could include a kinematic equation.) <u>Note:</u> Student cannot earn this point even if justification is consistent with an incorrect FBD in part (a)		1 point
For a logical, relevant, and internally consistent argument that addresses the required argument or question asked, and follows the guidelines described in the published requirements for the paragraph-length response		1 point
<u>Note:</u> A maximum of 4 of 5 points (first 3 points plus 5th point) can be earned if the FBDs in part (a) are incorrect (e.g., friction force is in the same direction in both FBDs)		

<i>Alternate solution using work/energy reasoning for 1st two points of part (b)</i>		
<i>For correctly applying work-energy to up/down ramp (or, indicating block-Earth system has the same potential energy at the beginning and end)</i>		1 point
<i>For correct treatment of energy loss to friction</i>		1 point

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Question 4 (continued)

Learning Objectives (LO)

LO 2.B.1.1: The student is able to apply $F = mg$ to calculate the gravitational force on an object with mass m in a gravitational field of strength g in the context of the effects of a net force on objects and systems. [See Science Practices 2.2 and 7.2]

LO 3.A.1.1: The student is able to express the motion of an object using narrative, mathematical, and graphical representations. [See Science Practices 1.5, 2.1, and 2.2]

LO 3.A.2.1: The student is able to represent forces in diagrams or mathematically using appropriately labeled vectors with magnitude, direction, and units during the analysis of a situation. [See Science Practice 1.1]

LO 3.B.1.1: The student is able to predict the motion of an object subject to forces exerted by several objects using an application of Newton's second law in a variety of physical situations with acceleration in one dimension. [See Science Practices 6.4 and 7.2]

LO 3.B.1.3: The student is able to reexpress a free-body diagram representation into a mathematical representation and solve the mathematical representation for the acceleration of the object. [See Science Practices 1.5 and 2.2]

LO 3.B.2.1: The student is able to create and use free-body diagrams to analyze physical situations to solve problems with motion qualitatively and quantitatively. [See Science Practices 1.1, 1.4, and 2.2]

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Question 5

7 points total

**Distribution
of points**

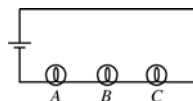


Figure 1

Students in a physics lab use a battery and three identical lightbulbs to construct the circuit shown in Figure 1 above. The brightness of each bulb increases when the current through the bulb increases.

(a) LO / SP: 5.C.3.1 / 6.4

1 point

Which bulb, if any, is the brightest?

 A *B* *C* All bulbs have the same brightness.

Briefly explain your reasoning in terms of current and/or potential difference.

Correct answer: “All bulbs have the same brightness”		
<u>Note</u> : The response does not earn credit if an incorrect selection is made.		
For correct checkbox and assertion that current (rate of charge flow) is the same throughout the circuit. <u>Note</u> : For this point to be earned, statement must include current and/or potential difference (i.e., “because they are in series” is not sufficient).		1 point
Examples: The current through the light bulbs is the same everywhere within a series circuit regardless of their proximity to the battery. Because the potential difference across each bulb is the same.		

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Question 5 (continued)

**Distribution
of points**

- (b) LO / SP: 5.B.9.3 / 2.2, 6.4; 5.C.3.1 / 6.4; 5.C.3.3 / 1.4, 2.2
3 points

The students observe the brightness of bulbs *A*, *B*, and *C*. The students then add a fourth lightbulb *D*, with a larger resistance than bulbs *A*, *B*, and *C*, as shown in Figure 2 below.

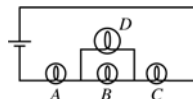


Figure 2

- i 2 points

Is bulb *A* now brighter than, dimmer than, or the same brightness as it was in the circuit in Figure 1 ?

____ Brighter ____ Dimmer ____ Same brightness

Briefly explain your reasoning.

Correct answer: “Brighter”		
<u>Note</u> : The response is graded even if an incorrect selection is made.		
For recognizing that the addition of lightbulb <i>D</i> reduces the equivalent resistance of the circuit		1 point
For correctly stating that the overall current (i.e., current through battery) is greater in the modified circuit		1 point

- ii 1 point

Is bulb *B* now brighter than, dimmer than, or the same brightness as it was in the circuit in Figure 1 ?

____ Brighter ____ Dimmer ____ Same brightness

Briefly explain your reasoning.

Correct answer: “Dimmer”

Note: The response does not earn credit if an incorrect selection is made.

For an argument that the current is now split between bulb <i>B</i> and bulb <i>D</i> , which tends to lower the current through bulb <i>B</i>		1 point
<u>Note</u> : It is not necessary to address change in current through the battery.		
<i>Alternate Solution</i>		
For an argument that the potential difference across both bulb <i>A</i> and bulb <i>C</i> increases <u>and</u> the potential difference of the battery remains constant, which causes the potential difference across bulb <i>B</i> to decrease due to conservation of energy (Kirchhoff’s Loop Rule)		1 point

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Question 5 (continued)

**Distribution
of points**

(c) LO / SP: 5.B.9.3 / 2.2, 6.4; 5.C.3.1 / 6.4; 5.C.3.3 / 1.4, 2.2
3 points

Bulbs *A*, *B*, and *C* each have resistance $10\ \Omega$, while bulb *D* has a resistance of $30\ \Omega$. In the circuit shown in Figure 2, the current through bulb *D* is measured to be $0.20\ \text{A}$. Calculate the potential difference across bulb *A*.

Reasoning using current:		
For correctly calculating the current through bulb <i>B</i> ($0.6\ \text{A}$)		1 point
For an implicit or explicit calculation of the total current consistent with previous step ($0.8\ \text{A}$)		1 point
For a potential difference across bulb <i>A</i> consistent with previous steps and with correct units: $8\ \text{V}$		1 point

Reasoning using potential difference:		
For correctly calculating the potential difference across bulb <i>D</i> and recognizing (explicitly or implicitly) that it is equal to the potential difference across bulb <i>B</i> ($6\ \text{V}$)		1 point
For explicitly or implicitly determining the ratio R_A/R_{BD} ($4/3$)		1 point
For a potential difference across bulb <i>A</i> consistent with previous steps and with correct units: $8\ \text{V}$		1 point

Learning Objectives (LO)

LO 5.B.9.3: The student is able to apply conservation of energy (Kirchhoff’s loop rule) in calculations involving the total electric potential difference for complete circuit loops with only a single battery and resistors in series and/or in, at most, one parallel branch. [See Science Practices 2.2, 6.4, 7.2]

LO 5.C.3.1: The student is able to apply conservation of electric charge (Kirchhoff’s junction rule) to the comparison of electric current in various segments of an electrical circuit with a single battery and resistors in series and in, at most, one parallel branch and predict how those values would change if configurations of the circuit are changed. [See Science Practices 6.4, 7.2]

LO 5.C.3.3: The student is able to use a description or schematic diagram of an electrical circuit to calculate unknown values of current in various segments or branches of the circuit. [See Science Practices 1.4, 2.2]